



Projected impacts of future climate change on the aboveground biomass of seagrasses at global scale

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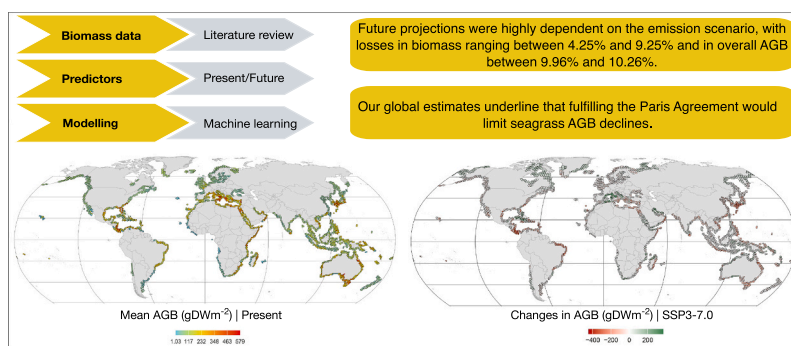
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HIGHLIGHTS

- Global AGB patterns are mainly defined by genus and temperature.
- Changes in global seagrass AGB were predicted under future scenarios.
- Future projections were highly dependent on the emission scenario.
- Average AGB losses range from 4.25 % to 9.25 %, and overall AGB from 9.96 % to 10.26 %.

GRAPHICAL ABSTRACT



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ABSTRACT

Seagrasses are crucial marine ecosystems that have experienced declines due to anthropogenic and climate change impacts. The projected future climate change suggests additional seagrass losses, but no global-scale estimates are currently available on the potential changes in aboveground biomass (AGB) of seagrasses. We modelled and quantified the current potential aboveground biomass (AGB) of seagrasses on the global scale and projected future AGB under contrasting Shared Socioeconomic Pathway (SSP) scenarios, from low emissions (SSP1–1.9) to high emissions (SSP3–7.0 and SSP5–8.5). A machine learning algorithm (Boosted Regression Trees) fitted a comprehensive AGB dataset against biological and anthropogenic meaningful predictors. The model performed with high accuracy (deviance explained: 0.83), highlighting the role of genus and temperature conditions in defining global AGB patterns. The model estimated a present-day average AGB of 133.83 gDW·m⁻² (DW, dry weight) and a total global AGB of 0.0673 Pg DW. Future projections were highly dependent on the emission scenario, with losses in AGB ranging between 4.25 % and 9.25 % and in overall AGB between 9.96 % and 10.26 % across scenarios. Particularly, the higher emission scenario projected severe regional losses along

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the coastlines of the Tropical Eastern Pacific, the Eastern Indo-Pacific, the Temperate Northern Pacific, and the Tropical Atlantic, and gains along the Temperate Southern Africa and the Arctic regions. Our global estimates underline that fulfilling the Paris Agreement, as well as conserving and monitoring populations most affected by combined anthropogenic pressures would help to limit seagrass AGB declines, thereby supporting the multiple ecological services of seagrasses.

1. Introduction

Climate change mitigation policies to meet the goals of the Paris Agreement (United Nations Framework Convention on Climate Change) include so-called blue carbon to conserve and restore coastal vegetated habitats and their disproportionately large carbon stocks (Duarte et al., 2013; IPCC, 2014; Mazarrasa et al., 2018). In this context, seagrasses are acknowledged as an important component of the marine carbon cycle with a significant role in climate change mitigation. Seagrasses also provide important habitats that stimulate biodiversity and support a wide range of additional ecological and socioeconomic services, including nutrient cycling, sediment stabilization, and coastal protection (Unsworth and Cullen-Unsworth, 2014). This places them among the most valuable ecosystems, with benefits on the orders of ~US\$ 2 trillion per year (Costanza et al., 2014; Nguyen et al., 2021). However, seagrasses have experienced global declines (de los Santos et al., 2019; Dunic et al., 2021; Tang and Hadibarata, 2022; Waycott et al., 2009) due to multiple anthropogenic pressures (Strydom et al., 2020) with associated losses in ecosystem functions and services (Dahl et al., 2023).

Seagrass aboveground biomass (AGB) is an important short-term organic carbon pool (Kennedy et al., 2022) that responds to the changing environment (Roca et al., 2016). Seagrass declines are often associated with conditions surpassing physiological thresholds (Fig. S1) (Edwards, 2021; Marbà et al., 2022), leading to impaired carbon balance (Marín-Guirao et al., 2016) and growth rates (Ondiviela et al., 2014; Serrano et al., 2021; Tang and Hadibarata, 2022). For instance, extreme thermal conditions (e.g., marine heatwave events) have led to seagrass AGB declines in Australia (Arias-Ortiz et al., 2018; Strydom et al., 2020), Southeast Asia, and the Caribbean (Steiner and Willette, 2015) and Mediterranean Seas (Marbà and Duarte, 2010). In addition to warming, excess nutrient levels, dredging, trawl fishing, changes in hydrodynamic energy and other physical and anthropogenic pressures, can further impact seagrass AGB (de Smit et al., 2022; Turschwell et al., 2021). Future climate projections suggest a risk of additional impacts on seagrass meadows (Fourqurean et al., 2012; Kwiatkowski et al., 2020; Marbà et al., 2022; Turschwell et al., 2021), yet no estimate is available on the potential changes in seagrass AGB at global scale. Previous studies projecting climate change impacts on seagrasses have e.g., focused on distribution range shifts in the Arctic (Assis et al., 2022) or at other regional scales (Chefaoui et al., 2018, 2021). While global analyses/models (Kennedy et al., 2022) and local maps (McHenry et al., 2023) of seagrass sediment carbon storage have recently been developed, no predictions of present-day global seagrass AGB are available considering key environmental and anthropogenic drivers. Such global seagrass AGB models would also have the potential to predict seagrass AGB for yet uncharted regions expected to harbor extensive seagrass meadows (Unsworth et al., 2019). Our models predict such baseline estimates of biomass for regions that have not yet been sampled.

Here we quantify present-day and end of the 21st century AGB of seagrasses by using machine learning modelling linking a comprehensive dataset of seagrass biomass (dry weight; DW) with biologically meaningful climatic predictors and global layers of anthropogenic pressure, a modelling approach recently implemented with high performance for seaweed productivity (Pessarrodona et al., 2022) and mangrove biomass at the global scale (Gouvêa et al., 2022). Models projecting future changes considered contrasting scenarios of greenhouse gas trajectories, namely the Shared Socioeconomic Pathway (SSP) “sustainability” scenario 1–1.9 in line with the Paris Agreement

expectations, and SSP3–7.0 and SSP5–8.5 scenarios representing increased emissions over time (Kwiatkowski et al., 2020). Our results, provided at global biogeographic and exclusive economic zone levels, may inform large-scale conservation, management actions and restoration programs, as well as updated estimates of the value of seagrass ecosystems.

2. Methods

The machine learning Boosted Regression Trees (BRT) algorithm was used to model seagrass AGB data against the set of biologically relevant predictor variables. A dataset of seagrass AGB (dry weight; DW) was acquired from the systematic review of Strydom et al. (2022) (Fig. S1; Fig. S2; Table S1), which comprehensively covers the available seagrass literature. Records were restricted to the period 2000–2020 to align with the timing of the predictor variables and to account for significant changes in occurrences caused by the combined effects of climate change and other pressures. The records were averaged whenever multiple records fell in the same cell (see below, Bio-ORACLE resolution of 0.05°, approximately 5 km). We assume that this step did not significantly impact biomass modelling since local variability (at the cell level) was low (1827 records of AGB averaged into 270 cells, with an average coefficient of variation of 0.79).

Relevant predictor variables for seagrasses (Fig. S3; (Gouvêa et al., 2024a,b)) were developed with the Bio-ORACLE computational pipelines (Assis et al., 2024); namely, sea benthic temperatures (long-term average of monthly maximum and minimum), nutrient conditions (nitrate), Light, wave energy (Fragkopoulou et al., 2022), salinity, and distance to river Delta (Gouvêa et al., 2022). Present-day data (decade 2000–2020) were acquired from the Copernicus Marine Environment Monitoring Service. End of century (decade 2090–2100) data were acquired from 11 Atmospheric-Ocean General Circulation Models (ACCESS-ESM1–5, CanESM5, CESM2-WACCM, CNRM-ESM2–1, GFDL-ESM4, GISS-E2–1-G, IPSL-CM6A-LR, MIROC-ES2L, MPI-ESM1–2-LR, MRI-ESM2–0, and UKESM1–0-LL) provided by the Coupled Model Intercomparison (CMIP) Project phase 6 (Earth System Grid Federation), under the scenarios SSP1–1.9, SSP3–7.0 and SSP5–8.5. Despite the relevance of additional predictors for seagrass AGB, e.g., the nature of seafloor (Fakiris et al., 2023), these was not considered because they are unavailable in CMIP.

The anthropogenic layers (Halpern et al., 2019) included in the models were selected based on pressures with significant impacts on seagrasses, as reported by Turschwell et al. (2021). These layers (from 2003 to 2013) encompassed shipping activities, destructive demersal commercial fishing, organic chemical pollution, extreme sea surface temperature events and turbidity. They were acquired from repository <https://kn.b.ecoinformatics.org/view/doi:10.5063/F12B8WBS>, except for extreme events of temperature and turbidity, for which we used maximum temperature and light as proxies, acquired from Bio-ORACLE. Anthropogenic pressure layers were kept constant in future projections due to the lack of available data. Projecting anthropogenic factors is challenging as they depend on dynamic socioeconomic conditions influenced by policies and global events.

The predictor variables were limited to the typical maximum depth range of seagrasses using the bathymetry (extending to a maximum of 20 m; (Larkum et al., 2018)). Prior to modelling, Pearson's correlation coefficient (Fig. S3) and variance inflation factor (VIF; Table S2) were estimated between predictors.

The BRT algorithm fitted AGB against the predictor variables using a Gaussian distribution. Hyperparameter tuning, a crucial step in machine learning algorithms was performed through spatial blocks cross-validation with 10 folds, to reduce overfitting (Roberts et al., 2017; Valavi et al., 2019). This step used the “grid search” method to test all hyperparameter combinations of the number of trees ranging between 50 and 1000 (step 50) of tree complexity between 1 and 6 (step 1), learning rate between 0.001 and 0.1 (step 0.005) and *n.minobsinnode* between 1 and 10 (step 1). To further reduce overfitting in the models, monotonicity constraints were implemented (Hofner et al., 2011), based on the expected biological outcome of each predictor variable on the response of the models as reported in the literature (Infantes et al., 2009; Krause-Jensen et al., 2003; Turschwell et al., 2021; Uhrin and Turner, 2018). Particularly, monotonic constraints ensure that as predictors change, the response variable consistently follows the same upward or downward trend, following biological expectations for each predictor.

In particular, positive for minimum temperature, salinity, Light and nutrient conditions, and negative for maximum temperature, wave energy, distance to river Delta, shipping, fishing and organic chemical pollution. Because seagrass taxa have different inherent AGB levels (Collier et al., 2021), a fixed effect factor was included in the model for “genus”, the lower taxonomic level available in the AGB dataset (Strydom et al., 2022). The performance of the models estimated in cross-validation was reported with deviance explained. The best hyperparameter combination (used in the final AGB model) was chosen as that maximizing the average deviance explained determined in cross-validation. The significance of the final model was explored by determining the relative contribution of each predictor variable in explaining AGB data variability and by developing partial dependency functions depicting the effect of each predictor on modelled AGB (Elith et al., 2008). Partial dependence plots (PDPs) illustrate how the biomass relates to one or more predictor variables, averaging the effects of other predictors. This highlights the marginal impact of a predictor on the response, offering insights into feature influence in complex models (Greenwell, 2017).

Seagrass AGB was predicted for present-day conditions and for the three SSP scenarios. Estimates were restricted to where seagrasses are known to occur by clipping predictions with range maps per genus obtained from the database of range maps of marine ecosystem structuring species (Gouvêa et al., 2024a). The estimates for present AGB were restricted to where seagrasses are known to occur, as well as the future projections were delimited to regions where seagrasses are projected to occur based on future range maps (Gouvêa et al., 2024a). This data paper provides all distribution maps by species.

Global estimates were aggregated as the maximum AGB per cell (to reduce the impact of overlapping low AGB estimates between genera) and reported as global maps and tables per Realm, ecoregion (Spalding et al., 2007) and Exclusive Economic Zone (EEZ). The modelling approach used for the range maps (Gouvêa et al., 2024a) was the same as that for the AGB estimates. The difference is the response variable, i.e., distribution modelling involved the presences and pseudoabsences, whereas biomass models included the AGB data compiled from Strydom et al. (2022).

Finally, to evaluate areas of extrapolation and interpolation of models, the Multivariate Environmental Similarity Surface (MESS) index (Zurell et al., 2012) was used under present-day conditions, SSP1–1.9, SSP3–7.0, and SSP5–8.5. This approach helps to identify regions where SDMs accurately interpolate, and conversely, areas where the models extrapolate beyond the environmental range of biomass records.

All analyses were performed in the R programming language (R Core Team, 2023). The additional data are openly available in Figshare at <https://doi.org/10.6084/m9.figshare.24989082>.

3. Results

The BRT algorithm fitting a comprehensive dataset of AGB (Figs. 1a and S2; Strydom et al., 2022) retrieved a deviance explained of 0.83 (Fig. 1a), a Person's Correlation of 0.91 and an average difference between predicted and observed AGB of $0.99 \pm 1.95 \text{ g DW.m}^{-2}$. The final combination of hyperparameter was tree complexity = 6, learning rate = 0.05, number of trees = 150 and *n.minobsinnode* = 2. Uncertainty, given by the standard deviation of the average AGB, was up to 0.18, particularly in the Arabic Sea region (Fig. S4). Despite the high accuracy of our model and the low uncertainty, extrapolation areas of biomass values were identified, particularly in regions with high AGB of seagrasses (Fig. S5). Northeastern Brazil, the Arctic, the Sunda Shelf/Java, the Southern Red Sea, the Arabian (Persian) Gulf, the Gulf of Oman, and the Southeast Australian Shelf may exhibit the highest uncertainty due to extrapolations.

Differences at the genus level largely explained the variability of global seagrass AGB (Fig. 1b; relative contribution of 34.87 %), followed by maximum temperature and shipping (contribution of 18.39 % and 11.39 %). Nutrient levels, salinity, and wave energy had similar lower contributions to the model (contributions ranging between 8.85 % and 6.47 %). Chemical organic pollution, minimum temperature, distance to river delta, fishing and light were the least important predictors to explain AGB data across areas (relative contribution < 5 %). The two temperature predictors (minimum and maximum) showed a strong correlation between them (Pearson's correlation 0.90; VIF > 5; Fig. S3; Table S2), yet their opposite monotonic fit removed any confounding effect regarding their contribution to the response of the model. No other predictors showed a strong correlation among them. The partial dependency functions showed particularly high effect on AGB for the genera *Amphibolis*, *Posidonia* and *Enhalus* and low effect on AGB for *Halodule* and *Halophila* (Fig. 2a), which reflect differences in genera-specific AGB (Table S1). In this way, the species characterized by the highest AGB per unit area had the most significant relative effect in the model response, while those with the lowest AGB exhibited the least effect (Table S1). These functions also showed, at global scale for community level, increased average AGB with increasing temperature between 25.53 and 29.77 °C (from minimum and maximum temperatures, respectively), shipping traffic < 0.01, nutrients above 0.75 mmol m^{-3} and salinity above 30.06 (Figs. 2b, S6 and S7).

Predicting for present-day conditions estimated an average AGB of $133.83 \pm 88.80 \text{ g DW.m}^{-2}$ and an overall AGB of 0.0673 Pg DW (Tables 1 and 2; Fig. 3a). At the biogeographic scale, the Tropical Atlantic, Temperate Northern Atlantic, and the Central and Western Indo-Pacific (Table 1) summed up to approximately 80 % of the overall seagrass biomass, with hotspots predicted in the ecoregions of Northeastern Brazil, Greater Antilles, Tunisian Plateau/Gulf of Sidra, Levantine Sea, the Western Mediterranean, the Southern and Southwestern Caribbean, the Southern Gulf of Mexico, the Sunda Shelf/Java Sea, the Eastern Philippines and Northern and Central Red Sea (regions with overall AGB > 0.0012 Pg DW; Table S3). At the EEZ level, hotspots were predicted in Australia, Indonesia, Brazil, United States, Mexico, Japan, China, Philippines, Indonesia, Italy, India, Somali, Malaysia, Madagascar, Egypt and Saudi Arabia (countries with overall AGB > 0.0012 Pg DW; Table S4).

Projecting for the two SSP scenarios estimated an average AGB of $128.14 \pm 75.10 \text{ g DW.m}^{-2}$ and $121.17 \pm 79.20 \text{ g DW.m}^{-2}$ by the end of the Century (Tables 1 and 2; Fig. 3b and c) corresponding to global biomass declines of 4.25 and 9.25 % (SSP1–1.9 and SSP3–7.0, respectively) when compared to the present (Table 1). The projections of overall AGB ranged between 0.0606 and 0.0604 Pg DW, corresponding to decreases of 9.96 and 10.26 % (SSP1–1.9 and SSP3–7.0, respectively) relative to present-day estimates. Losses at the biogeographic level were mostly projected in the Tropical Eastern Pacific, the Eastern Indo-Pacific, the Tropical Atlantic and the Temperate Northern Pacific, more specifically, in the ecoregions of Nicoya, Northeastern Brazil,

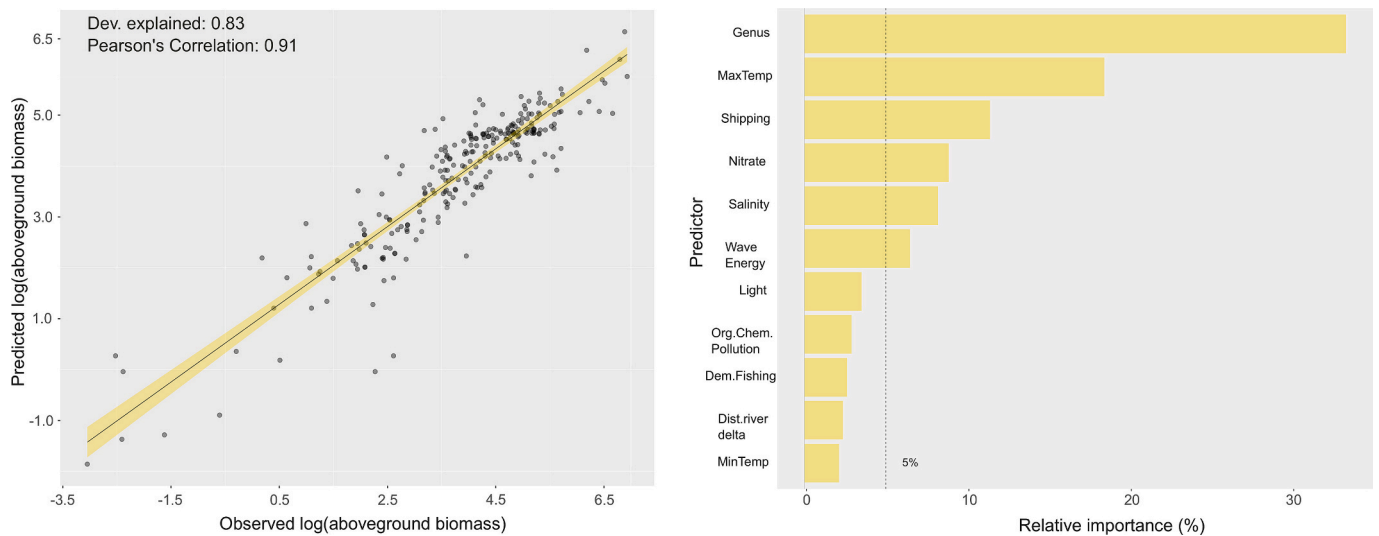


Fig. 1. (a) Relationship between observed and predicted seagrass biomass (natural logarithm of aboveground biomass as $\text{gDW} \cdot \text{m}^{-2}$; Deviance explained: 0.83, $p < 0.0001$), and (b) Relative contribution (%) of predictor variables to the AGB model. Dotted lines depict a contribution threshold of 5 %. Max and min. Temperature ($^{\circ}\text{C}$), Shipping activities measured as the maximum number of shipping tracks recorded in a grid cell, nitrate ($\text{mmol} \cdot \text{m}^{-3}$), wave energy ($\text{m} \cdot \text{s}^{-1}$), Org. chemical pollution (model based on the application of pesticides), demersal fishing (based on annual wild-caught industrial fisheries catch for trawl and dredge fisheries) and distance to river delta (km).

Panama Bight, Yellow Sea, Northern Bay of Bengal, Central Kuroshio Current (regions with >70 % of losses in the overall AGB; Table 1; Table S3), with risk of species extinction in Amazonia, Society Islands and Gulf of Guinea. At the country level, it is important to highlight that Brazil, Costa Rica, Panama, Japan and China, predicted to have higher present-day overall biomass, are expected to be severely impacted under the SSP3–7.0 (losses >60 %; Table S4). Despite the general trend of losses, pronounced increases in overall biomass were projected in Temperate southern Africa and the Arctic realms (by up to 211.22 % in the SSP3–7.0; Table 1).

Additionally, contrasting trends were also projected for specific regions. For instance, Arctic, Western Indo-Pacific, Temperate South America and Northern Atlantic were projected to lose or gain average AGB, but the opposite was projected when considering the overall AGB.

Comprehensive estimates per ecoregion and Exclusive Economic Zone are available in Tables S3 and S4 and future projections of seagrass AGB under SSP5–8.5 are provided in Fig. S8.

4. Discussion

Machine learning fitting comprehensive biomass, climate and anthropogenic datasets provided new estimates of global seagrass AGB for present-day and for the end of the 21st century conditions. On the global scale, modelling quantified an overall AGB of 0.0673 Pg DW , which can be impacted under future climate, however with changes particularly dependent on the emission scenario considered. By exceeding the climate forcing underlying the Paris Agreement expectations, global AGB losses of up to 10.26 % are projected, with more severe impacts along tropical and warm-temperate regions. Countries like Brazil, Costa Rica, Panama, Japan and China currently predicted with higher AGB might suffer marked changes in seagrasses biomass and associated ecosystem services, adding up to the losses already documented in the past (Orth et al., 2006; Short et al., 2006, 2011). Conversely, predicted increases in AGB along the Arctic and the Temperate southern African coast may partly counteract the general trend of losses, with regional gains in AGB by up to 211.22 %. Overall, our impact assessment underlines the substantial benefits fulfillment of the Paris Agreement would have for seagrass meadows, reducing projected end-of-century declines. The information provided at the biogeographic and EEZ levels can be used to inform seagrass

conservation and management actions (Wedding et al., 2021).

The machine learning model achieved high performance (83 % of variability explained) and low uncertainty, with the contribution and effect of predictor variables for seagrass AGB largely matching those of previous studies of drivers of seagrass distribution (Jayatilake and Costello, 2018; Turschwell et al., 2021). Despite the overall agreement, variations between estimates, projections and variable contributions are expected due to different methods and predictors used in the models.

The most important predictor explaining the variability of AGB across the present global distribution of seagrasses was the seagrass genus, which reflects inherent traits of the modelled genera (Fig. 2). The relative effect of the specific genus largely matched available empirical seagrass genera AGB (Table S1), with the highest relative effect on AGB for *Amphibolis*, *Posidonia* and *Enhalus*, and the lowest ones for *Halodule* and *Halophila* (Fig. 2) (Collier et al., 2021). Seagrass traits were also identified, together with geomorphological settings, as key predictors of soil carbon stocks of seagrass meadows (Kennedy et al., 2022).

Thermal conditions were also important in explaining the variations in seagrass AGB across the global dataset with predicted optima found by up to 30°C , but again, highly dependent on the genus considered. Accordingly, the dataset used to fit the model showed a maximum observed temperature per genus ranging between 26.52°C and 32°C , which translated into high variability in observed biomass for such temperatures ranging between $33.63 \text{ g} \cdot \text{m}^{-2}$ (*Cymodocea*) and $204.76 \text{ g} \cdot \text{m}^{-2}$ (*Posidonia*) (Table S1). Such variability across taxa is further demonstrated in additional experimental assessments that show seagrass tolerances ranging between 25°C and 38°C (Marbà et al., 2022; Fig. S1b). While some seagrasses thrive in the warm conditions of the Red Sea and the Arabian Gulf, where temperatures can easily reach 35°C during summer (Marbà et al., 2022), temperate species, such as *Zostera marina*, have much lower temperature tolerances of up to 23.5°C (Beca-Carretero et al., 2018; Lee et al., 2007), a threshold that is slightly below the limits of our model (minimum temperature from 25.53°C ; Fig. 2). The fixed effect placed on genera in the models aimed to account for such variability.

Declines in seagrass meadows have also been associated with higher pressures from shipping traffic (Turschwell et al., 2021). Such pressure can elevate the risk of oil spills, while dredging and associated sediment resuspension around ports and along shipping channels can impact the survival of seagrass meadows (Grech et al., 2012). Shipping pressures

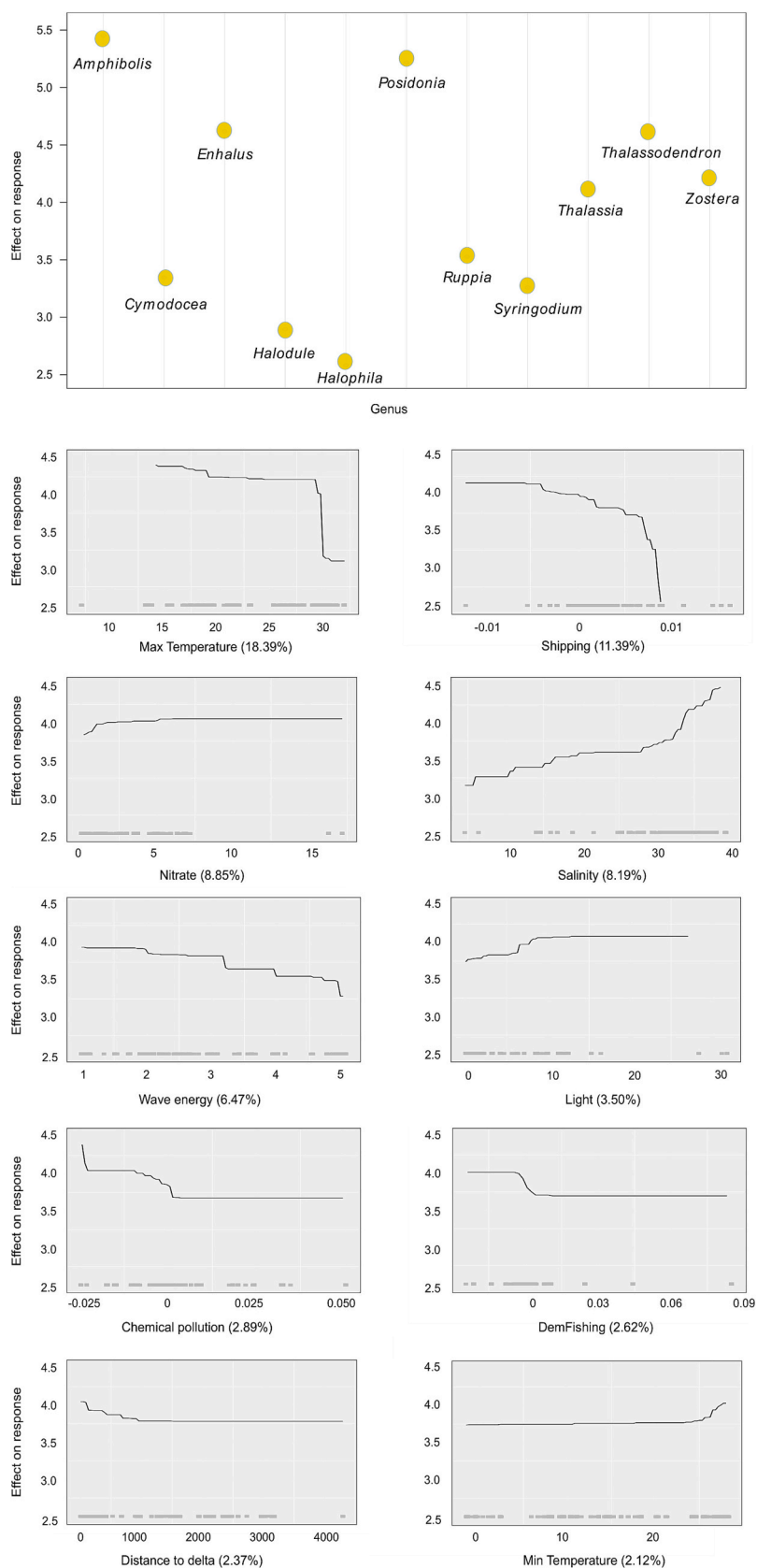


Fig. 2. Partial dependency functions (with monotonicity constraints) depicting the effect and relative contribution (%) of (a) genus and (b) climate and anthropogenic predictors on the response of the AGB model. Gray square dots along the x-axis depict the climatic regions where biomass was observed.

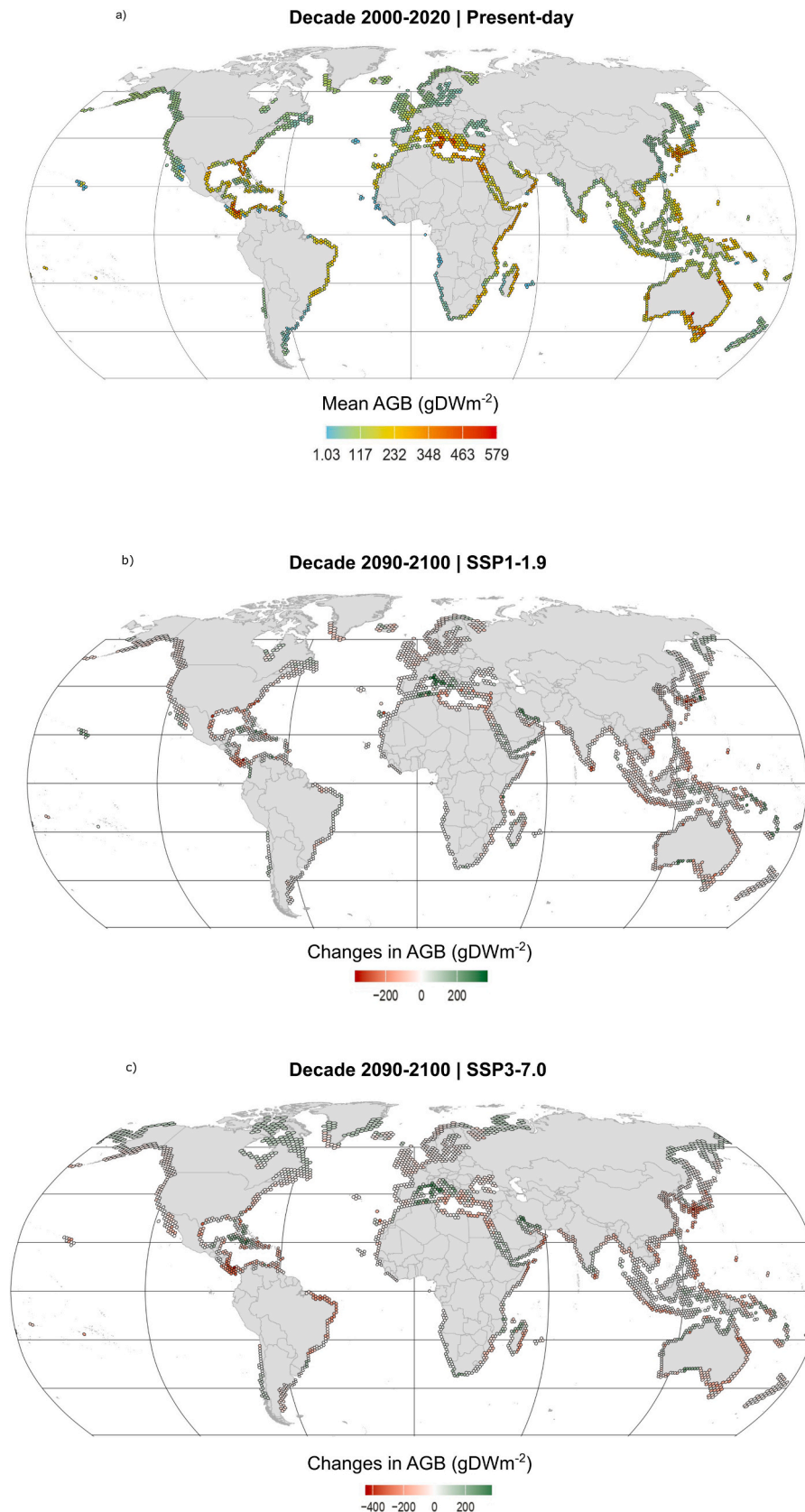


Fig. 3. Potential distribution of seagrass aboveground biomass (a) predicted for present-day (decade 2000–2020) and projected changes for end of 21st century conditions under the (b) SSP1–1.9 and (c) SSP 3–7.0 scenarios of future climate change. Maps were aggregated equal-size polygons for better visualization. Future projections for SSP5–8.5 are provided in Fig. S8. Present-day estimates of AGB were restricted to seagrass distribution range maps for current conditions, and future estimates of AGB were restricted to range maps representing future seagrass distribution projections. Refer extrapolation areas and uncertainties inferred by the model in the supplement (Figs. S4 and S5).

Table 1

Potential average (gDW-m⁻²) and overall aboveground biomass (Pg) of seagrass meadows predicted per marine realm (according to Spalding et al., 2007) for present-day (decade 2000–2020) and the end of 21st century conditions under the (b) SSP1–1.9 and (c) SSP 3–7.0 scenarios of future climate change. Comprehensive estimates per ecoregion and Exclusive Economic Zones are available in Tables S3 and S4.

Realm	Average biomass (gDW-m ⁻²)			Overall biomass (Pg)		
	Present	Changes (%)		Present	Changes (%)	
		SSP 1–1.9	SSP 3–7.0		SSP 1–1.9	SSP 3–7.0
Tropical Atlantic	163.79	12.27	–5.82	0.0094	–6.34	–35.79
Tropical Eastern Pacific	192.36	–51.85	–86.16	0.0006	–36.58	–90.36
Central Indo-Pacific	135.68	–14.80	–5.03	0.0178	–16.03	–2.83
Eastern Indo-Pacific	112.62	54.83	–36.74	0.0001	28.67	–58.48
Western Indo-Pacific	158.19	–2.75	10.59	0.0135	–4.94	–2.56
Temperate Australasia	192.56	–2.86	–12.21	0.0053	–10.53	–15.80
Temperate Northern Atlantic	125.66	4.56	8.58	0.0137	–7.68	–5.62
Temperate Southern Africa	81.92	8.43	60.03	0.0004	0.73	48.22
Temperate South America	72.62	–3.91	–19.74	0.0006	5.76	1.22
Temperate Northern Pacific	103.72	–12.03	–30.45	0.0049	–16.13	–35.77
Arctic	91.72	–11.84	–19.21	0.0005	–9.93	211.22

Table 2

Potential area (km²; Gouvêa et al., 2024b), average (g DW-m⁻²) and overall aboveground biomass (Pg DW) of seagrasses meadows predicted at global scales for present-day (decade 2000–2020) and the end of 21st century conditions under the (b) SSP1–1.9 and (c) SSP3–7.0 scenarios of future climate change.

Scenario	Area (km ²)	Mean AGB (gDW m ⁻²)	Changes (%)	Overall AGB (Pg)	Changes (%)
Present-day	917,169	133.83		0.0673	
SSP1–1.9	867,244	128.14	–4.25	0.0606	–9.96
SSP3–7.0	970,902	121.17	–9.25	0.0604	–10.26

are expected to intensify in the future driven by Maritime Belt and Road Initiative, which will likely lead to a rise in global shipping traffic (McMahon et al., 2022; Turschwell et al., 2021). In this context, human activities do not impact all seagrass species equally because those with different life-history strategies respond differently to extreme conditions. For instance, species with colonizing characteristics, such as *Halophila* spp., known for their rapid shoot turnover, are generally more capable of recovering from disturbances. In contrast, persistent species like *Posidonia* spp. are long-lived, resistant to pressures but are slower to recover (Turschwell et al., 2021). Although persistent seagrass species are distributed more narrowly worldwide, they are considered to provide more diverse services than colonizing species in the areas where they exist (Mtwana Nordlund et al., 2016).

The predictors explaining the remaining part of AGB data are also supported by previous studies linking seagrass growth with non-limiting nutrient conditions, in particular, nitrate concentrations (Mazarrasa et al., 2018; Touchette and Burkholder, 2000) above approx. 1 mmol m⁻³ (Gouvêa et al., 2024b) and salinity above 30 (Touchette, 2007;

Gouvêa et al., 2024b). However, while nutrient concentrations above a minimum are needed to support seagrass growth, excess nutrient concentration reflecting eutrophication is a major stressor on seagrass meadows, although this is not directly reflected in the overall model predictions (Fig. 2). For example, a century long dataset from the Baltic/North Atlantic region identified eutrophication as a major driver of large-scale seagrass declines. It underlined the need for holistic management of the multiple stressors to increase seagrass resilience, which is increasingly important in the face of warming (Krause-Jensen et al., 2021).

The model allowed quantifying seagrass AGB at the global scale, with predictions pointing to a present-day average AGB of 133.83 ± 88.80 g.m⁻² and an overall AGB of 0.067 Pg DW, when integrating over the global extent of 917,619 Km² (Gouvêa et al., 2024a). The average AGB estimate is supported by the average of observed records used in the model (157.41 ± 111.98 g DW-m⁻²; Table S1), while the overall AGB is supported by a previous estimate ranging between 0.07 and 0.15 Pg (Fourqurean et al., 2012) in the same order of magnitude, based on multiplying the average of biomass in situ records by the global extent of seagrasses (Duarte and Chiscano, 1999; Fourqurean et al., 2012).

The distribution of predicted current AGB varied significantly, with global hotspots located in the Temperate Northern Atlantic, the Tropical Atlantic and the Central and Western Indo-Pacific (summing up to 80 % of the overall AGB; note that overall AGB estimates are dependent on the actual extent of EEZs). Particularly, Australia, Indonesia, Brazil, United States, Mexico, Japan, China, Philippines, Indonesia, Italy, India, Somali, Malaysia and Madagascar are predicted to contribute more to the global biomass of seagrasses and can guide conservation strategies with global significance.

Future projections of AGB were mainly dependent on the SSP scenario considered. This is somewhat expected, as contrasting scenarios of global change (e.g., SSP1–1.9 and SSP3–7.0) lead to different impacts on marine biodiversity (Assis et al., 2017; Assis et al., 2024; Gouvêa et al., 2022). For seagrasses, climate forcing beyond the Paris Agreement expectations (Marbà et al., 2022) may push overall AGB losses up to 10.26 % on the global scale, with major losses in the Tropical Eastern Pacific, the Eastern Indo-Pacific, the Tropical Atlantic and the Temperate Northern Pacific, which add to the already reported declines (e.g., MHW; Arias-Ortiz et al., 2018; Strydom et al., 2020). Future projections of aboveground biomass (AGB) for various marine mangrove species indicate poleward shifts into temperate regions (Coldren et al., 2019; Gouvêa et al., 2022; Osland et al., 2019). Additionally, in the coastlines of the Eastern Indo-Pacific and Tropical Eastern Pacific (Table 1), the trend of declining mangroves and seagrass meadows has been projected, with overall biomass losses surpassing 50 % (Gouvêa et al., 2022).

The risk of population declines in numerous regions by the end of the 21st century (Gouvêa et al., 2024b) and extinctions (e.g., Amazonia) might strongly disrupt the ecosystem services supported by seagrass, including carbon sequestration. Concomitantly, CO₂ emissions can be enhanced, a process linked to soil erosion where seagrass meadows are lost (Arias-Ortiz et al., 2018; Lovelock et al., 2017). The regional loss of seagrasses can further reduce the overall biomass and diversity of numerous associated species (O'Leary et al., 2021), some of which with high conservation and commercial value, the protection of shorelines (James et al., 2023), and ultimately, may lead to the erosion of seagrass genetic diversity levels and therefore species' evolutionary potential (Unsworth et al., 2019). Accordingly, seagrass conservation should be enhanced across the globe, with the meadows in Brazil, Costa Rica, Panama, Japan and China, where biomass losses were mostly projected. Despite the overall pattern of AGB losses, expansions may occur e.g., in the temperate southern Africa and along the Arctic, owing to poleward range shifts (Gouvêa et al., 2024b), leading to an enhancement of the overall AGB (Table 1). This is in line with previous studies describing future conditions to better support seagrass meadows in these areas (Assis et al., 2022; Krause-Jensen et al., 2020). On one side, the borealization of some species in such regions comes at the expense of losses

of native ones, with cold-tolerant species outcompeting with native species and altering the community compositions (Chan et al., 2019), following the trend documented along the past century.

The model achieved high performance and significance for explaining the current distribution of AGB observations across the globe; however, we must recognize some limitations that add uncertainties to AGB estimates for both current and future scenarios. First, the model estimated present-day and future AGB based on a limited set of climatic predictors, but additional drivers may synergistically explain the remaining variability of data (deviance explained of 0.83). For instance, extreme climatic events beyond long-term climatologies (e.g., marine heat waves) are relevant drivers of change in seagrass areas (Bryars et al., 2011; Dewsbury et al., 2016) but are not embedded in the models. Second, the lack of data on the nature of the seafloor at the global scale (Jayathilake and Costello, 2020) produced predictions that cover broader areas than those suitable to support seagrass growth, therefore, the reported estimates define the upper bounds of potential overall AGB. Third, estimating the impacts of climate change on biodiversity is challenging due to the lack of observations supporting projections (Elith et al., 2008). In particular, there is an inherent risk of extrapolating to conditions beyond the ranges covered by the data used to fit the model. Consequently, regions such as Northeastern Brazil, the Arctic, the Sunda Shelf/Java, the Southern Red Sea, the Arabian (Persian) Gulf, the Gulf of Oman, and the Southeast Australian Shelf may exhibit the highest uncertainty due to extrapolation in all climatic scenarios (Fig. S4 and S5). There is also uncertainty added by sea level rise estimates and the likely onshore migration of seagrass, as previously discussed elsewhere for mangrove forests (Schuerch et al., 2018).

5. Conclusions

Overall, our quantification of current- and projected future seagrass AGB at the global scale underlines the substantial benefits fulfillment of the Paris Agreement would have for the conservation of seagrass meadows and the maintenance of important associated ecosystem services (Johannessen, 2022). While the protection of seagrasses should be prioritized across the globe, our model identified biogeographic regions and countries particularly vulnerable to climate-related losses. This underscores the critical need for implementing management strategies that can alleviate these combined pressures and enhance the overall resilience of these meadows. Seagrass conservation and restoration can be integrated into National Blue Carbon Policy Assessments (Herr et al., 2016; Taillardat et al., 2018), e.g., through the establishment of marine protected areas, as evidence shows reduced population declines and increased carbon sequestration potential in protected areas (Jankowska et al., 2022). Such sustainable management of seagrass habitats would also stimulate marine biodiversity and thereby support the Kunming-Montreal Global Biodiversity framework (Obura et al., 2023; Fu et al., 2024).

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2025.178680>.

CRedit authorship contribution statement

Lidiane P. Gouvêa: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dorte Krause-Jensen:** Writing – review & editing, Visualization, Validation. **Carlos M. Duarte:** Writing – review & editing, Visualization, Validation. **Jorge Assis:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data and codes used in the aboveground biomass models of seagrasses for present-day conditions and future climate change are available at: <https://doi.org/10.6084/m9.figshare.24989082>.

References

- Arias-Ortiz, A., Serrano, O., Masqué, P., Lavery, P.S., Mueller, U., Kendrick, G.A., Rozaimi, M., Esteban, A., Fourqurean, J.W., Marbà, N., Mateo, M.A., Murray, K., Rule, M.J., Duarte, C.M., 2018. A marine heatwave drives massive losses from the world's largest seagrass carbon stocks. *Nat. Clim. Chang.* 8 (4), 338–344. <https://doi.org/10.1038/s41558-018-0096-y>.
- Assis, J., Bercebar, E., Claro, B., Alberto, F., Reed, D., Raimondi, P., Serrão, E.A., 2017. Major shifts at the range edge of marine forests: the combined effects of climate changes and limited dispersal. *Sci. Rep.* 7 (February), 1–10. <https://doi.org/10.1038/srep44348>.
- Assis, Jorge, Serrão, E.A., Duarte, C.M., Fragkopoulou, E., Krause-Jensen, D., 2022. Major expansion of marine forests in a warmer Arctic. *Front. Mar. Sci.* 9 (March), 1–10. <https://doi.org/10.3389/fmars.2022.850368>.
- Assis, Jorge, Fragkopoulou, E., Gouvêa, L., Araújo, M.B., Serrão, E.A., 2024. Kelp forest diversity under projected end-of-century climate change. *Divers. Distrib.* 30 (6), 1–11. <https://doi.org/10.1111/ddi.13837>.
- Beca-Carretero, P., Olesen, B., Marbà, N., Krause-Jensen, D., 2018. Response to experimental warming in northern eelgrass populations: comparison across a range of temperature adaptations. *Mar. Ecol. Prog. Ser.* 589. <https://doi.org/10.3354/meps12439>.
- Bryars, S., Collings, G., Miller, D., 2011. Nutrient exposure causes epiphytic changes and coincident declines in two temperate Australian seagrasses. *Mar. Ecol. Prog. Ser.* 441. <https://doi.org/10.3354/meps09384>.
- Chan, F. T., Stanislawczyk, K., Sneekes, A. C., Dvoretzky, A., Gollasch, S., Minchin, D., David, M., Jelmer, A., Albrechtsen, J., & Bailey, S. A. (2019). Climate change opens new frontiers for marine species in the Arctic: current trends and future invasion risks. In *Glob. Chang. Biol.* (Vol. 25, issue 1). doi:<https://doi.org/10.1111/gcb.14469>.
- Chefaoui, R.M., Duarte, C.M., Serrão, E.A., 2018. Dramatic loss of seagrass habitat under projected climate change in the Mediterranean Sea. *Glob. Chang. Biol.* 24 (10). <https://doi.org/10.1111/gcb.14401>.
- Chefaoui, R.M., Duarte, C.M., Tavares, A.I., Frade, D.G., Sidi Cheikh, M.A., Abdoull Ba, M., Serrão, E.A., 2021. Predicted regime shift in the seagrass ecosystem of the Gulf of Arguin driven by climate change. *Glob. Ecol. Conserv.* 32. <https://doi.org/10.1016/j.gecco.2021.e01890>.
- Coldren, G.A., Langley, J.A., Feller, I.C., Chapman, S.K., 2019. Warming accelerates mangrove expansion and surface elevation gain in a subtropical wetland. *J. Ecol.* 107 (1), 79–90. <https://doi.org/10.1111/1365-2745.13049>.
- Collier, C.J., Langlois, L.M., McMahon, K.M., Udy, J., Rasheed, M., Lawrence, E., Carter, A.B., Fraser, M.W., McKenzie, L.J., 2021. What lies beneath: predicting seagrass below-ground biomass from above-ground biomass, environmental conditions and seagrass community composition. *Ecol. Indic.* 121. <https://doi.org/10.1016/j.ecolind.2020.107156>.
- Costanza, R., de Groot, R., Sutton, P., van der Ploeg, S., Anderson, S.J., Kubiszewski, I., Farber, S., Turner, R.K., 2014. Changes in the global value of ecosystem services. *Glob. Environ. Chang.* 26 (1), 152–158. <https://doi.org/10.1016/j.gloenvcha.2014.04.002>.
- Dahl, M., McMahon, K., Lavery, P.S., Hamilton, S.H., Lovelock, C.E., Serrano, O., 2023. Ranking the risk of CO2 emissions from seagrass soil carbon stocks under global change threats. *Glob. Environ. Chang.* 78 (March 2022), 102632. <https://doi.org/10.1016/j.gloenvcha.2022.102632>.
- de los Santos, C.B., Krause-Jensen, D., Alcoverro, T., Marbà, N., Duarte, C.M., van Katwijk, M.M., Pérez, M., Romero, J., Sánchez-Lizaso, J.L., Roca, G., Jankowska, E., Pérez-Lloréns, J.L., Fournier, J., Montefalcone, M., Pergent, G., Ruiz, J.M.,

- Cabaço, S., Cook, K., Wilkes, R.J., Santos, R., 2019. Recent trend reversal for declining European seagrass meadows. *Nat. Commun.* 10 (1). <https://doi.org/10.1038/s41467-019-11340-4>.
- de Smit, J.C., Bin Mohd Noor, M.S., Infantes, E., Bouma, T.J., 2022. Wind exposure and sediment type determine the resilience and response of seagrass meadows to climate change. *Limnol. Oceanogr.* 67 (S1), S121–S132. <https://doi.org/10.1002/lno.11865>.
- Dewsbury, B.M., Bhat, M., Fourqurean, J.W., 2016. A review of seagrass economic valuations: gaps and progress in valuation approaches. *Ecosyst. Serv.* 18. <https://doi.org/10.1016/j.ecoser.2016.02.010>.
- Duarte, C.M., Chiscano, C.L., 1999. Aquatic I) otany seagrass biomass and production: a reassessment. *Aquat. Bot.* 65, 159–174.
- Duarte, C. M., Losada, I. J., Hendriks, I. E., Mazarrasa, I., & Marbà, N. (2013). The role of coastal plant communities for climate change mitigation and adaptation. In *Nat. Clim. Chang.* (Vol. 3, issue 11). doi:<https://doi.org/10.1038/nclimate1970>.
- Dunic, J.C., Brown, C.J., Connolly, R.M., Turschwell, M.P., Côté, I.M., 2021. Long-term declines and recovery of meadow area across the world's seagrass bioregions. *Glob. Chang. Biol.* 27 (17), 4096–4109. <https://doi.org/10.1111/gcb.15684>.
- Edwards, A.J., 2021. Impact of climatic change on coral reefs, mangroves, and tropical seagrass ecosystems. *Climate Change May*, 209–234. <https://doi.org/10.1201/9781003069935-10>.
- Elith, J., Leathwick, J.R., Hastie, T., 2008. A working guide to boosted regression trees. *J. Anim. Ecol.* 77 (4), 802–813. <https://doi.org/10.1111/j.1365-2656.2008.01390.x>.
- Fakiris, E., Giannakopoulos, V., Leftheriotis, G., Dimas, A., Papatheodorou, G., 2023. Predictive mapping of Mediterranean seagrasses-exploring the influence of seafloor light and wave energy on their fine-scale spatial variability. *Remote Sens.* 15 (11). <https://doi.org/10.3390/rs15112943>.
- Fourqurean, J.W., Duarte, C.M., Kennedy, H., Marbà, N., Holmer, M., Mateo, M.A., Apostolaki, E.T., Kendrick, G.A., Krause-Jensen, D., McGlathery, K.J., Serrano, O., 2012. Seagrass ecosystems as a globally significant carbon stock. *Nat. Geosci.* 5 (7), 505–509. <https://doi.org/10.1038/ngeo1477>.
- Fragkopoulou, E., Serrão, E.A., De Clerck, O., Costello, M.J., Araújo, M.B., Duarte, C.M., Krause-Jensen, D., Assis, J., 2022. Global biodiversity patterns of marine forests of brown macroalgae. *Glob. Ecol. Biogeogr.* 31 (4). <https://doi.org/10.1111/gcb.13450>.
- Fu, C., Steckbauer, A., Mann, H., Duarte, C.M., 2024. Achieving the Kunming–Montreal global biodiversity targets for blue carbon ecosystems. *Nat. Rev. Earth Environ.* 5 (7), 538–552.
- Gouvêa, L., Fragkopoulou, E., Legrand, T., Serrão, E.A., Assis, J., 2024a. Range map data of marine ecosystem structuring species under global climate change. *Data Brief* 52, 110023.
- Gouvêa, L., Fragkopoulou, E.B., Araújo, M., Serrão, E.A., Assis, J., 2024b. Seagrass biodiversity under the latest-generation scenarios of projected climate change. *J. Biogeogr.* 52 (1), 172–185.
- Gouvêa, L.P., Serrão, E.A., Cavanaugh, K., Gurgel, C.F.D., Horta, P.A., Assis, J., 2022. Global impacts of projected climate changes on the extent and aboveground biomass of mangrove forests. *Divers. Distrib.* 28 (11), 2349–2360. <https://doi.org/10.1111/ddi.13631>.
- Grech, A., Chartrand-Miller, K., Erfteimeijer, P., Fonseca, M., McKenzie, L., Rasheed, M., Taylor, H., Coles, R., 2012. A comparison of threats, vulnerabilities and management approaches in global seagrass bioregions. *Environ. Res. Lett.* 7 (2). <https://doi.org/10.1088/1748-9326/7/2/024006>.
- Greenwell, B.M., 2017. Pdp: an R package for constructing partial dependence plots. *R. J.* 9 (1), 421–436. <https://doi.org/10.32614/rj-2017-016>.
- Halpern, B.S., Frazier, M., Afflerbach, J., Lowndes, J.S., Micheli, F., O'Hara, C., Scarborough, C., Selkoe, K.A., 2019. Recent pace of change in human impact on the world's ocean. *Sci. Rep.* 9 (1), 1–8. <https://doi.org/10.1038/s41598-019-47201-9>.
- Herr, D., Himes-Cornell, A., Laffoley, D., 2016. *National Blue Carbon Policy Assessment Framework: Towards Effective Management of Coastal Carbon Ecosystems*. IUCN, Gland, Switzerland.
- Hofner, B., Muller, J., Hothorn, T., 2011. Monotonicity-constrained species distribution models, 92 (may 1895), 1895–1901.
- Infantes, E., Terrados, J., Orfila, A., Cañellas, B., Álvarez-Ellacuría, A., 2009. Wave energy and the upper depth limit distribution of *Posidonia oceanica*. *Bot. Mar.* 52 (5). <https://doi.org/10.1515/BOT.2009.050>.
- IPCC, 2014. 2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands, Published: IPCC, Switzerland. In: 2013 Supplement To the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands (Vol. 4, Issue October).
- James, R.K., Keyzer, L.M., van de Velde, S.J., Herman, P.M.J., van Katwijk, M.M., Bouma, T.J., 2023. Climate change mitigation by coral reefs and seagrass beds at risk: how global change compromises coastal ecosystem services. *Sci. Total Environ.* 857. <https://doi.org/10.1016/j.scitotenv.2022.159576>.
- Jankowska, E., Pelc, R., Alvarez, J., Mehra, M., Frischmann, C.J., 2022. Climate benefits from establishing marine protected areas targeted at blue carbon solutions. *Proc. Natl. Acad. Sci. USA* 119 (23). <https://doi.org/10.1073/pnas.2121705119>.
- Jayatilake, D.R.M., Costello, M.J., 2018. A modelled global distribution of the seagrass biome. *Biol. Conserv.* 226. <https://doi.org/10.1016/j.biocon.2018.07.009>.
- Johannessen, S.C., 2022. How can blue carbon burial in seagrass meadows increase long-term, net sequestration of carbon? A critical review. *Environ. Res. Lett.* 17 (9). <https://doi.org/10.1088/1748-9326/ac8ab4>.
- Kennedy, H., Pagès, J.F., Lagomasino, D., Arias-Ortiz, A., Colarusso, P., Fourqurean, J. W., Githaiga, M.N., Howard, J.L., Krause-Jensen, D., Kuwae, T., Lavery, P.S., Macreadie, P.I., Marbà, N., Masqué, P., Mazarrasa, I., Miyajima, T., Serrano, O., Duarte, C.M., 2022. Species traits and geomorphic setting as drivers of global soil carbon stocks in seagrass meadows. *Glob. Biogeochem. Cycles* 36 (10). <https://doi.org/10.1029/2022GB007481>.
- Krause-Jensen, D., Pedersen, M.F., Jensen, C., 2003. Regulation of eelgrass (*Zostera marina*) cover along depth gradients in Danish coastal waters. *Estuaries* 26 (4), 866–877. <https://doi.org/10.1007/BF02803345>.
- Krause-Jensen, D., Archambault, P., Assis, J., Bartsch, I., Bischof, K., Filbee-Dexter, K., Duntton, K.H., Maximova, O., Ragnarsdóttir, S.B., Sejr, M.K., Simakova, U., Spiridonov, V., Wegeberg, S., Winding, M.H.S., Duarte, C.M., 2020. Imprint of climate change on Pan-Arctic marine vegetation. *Front. Mar. Sci.* 7 (December), 1–28. <https://doi.org/10.3389/fmars.2020.617324>.
- Krause-Jensen, D., Duarte, C.M., Sand-Jensen, K., Carstensen, J., 2021. Century-long records reveal shifting challenges to seagrass recovery. *Glob. Chang. Biol.* 27 (3). <https://doi.org/10.1111/gcb.15440>.
- Kwiatkowski, L., Torres, O., Bopp, L., Aumont, O., Chamberlain, M., R. Christian, J., J. P. Dunne, Gehlen, M., Ilyina, T., G. John, J., Lenton, A., Li, H., S. Lovenduski, N., C. Orr, J., Palmieri, J., Santana-Falcón, Y., Schwing, J., Séférian, R., A. Stock, C., ... Ziehn, T. (2020). Twenty-first century ocean warming, acidification, deoxygenation, and upper-ocean nutrient and primary production decline from CMIP6 model projections. *Biogeosciences*, 17(13). doi:<https://doi.org/10.5194/bg-17-3439-2020>.
- Larkum, A.W.D., Kendrick, G.A., Ralph, P.J., 2018. *Seagrasses of Australia: Structure, Ecology and Conservation*. In: *Seagrasses of Australia: Structure, Ecology and Conservation*. Springer International Publishing. <https://doi.org/10.1007/978-3-319-71354-0>.
- Lee, K.S., Park, S.R., Kim, Y.K., 2007. Effects of irradiance, temperature, and nutrients on growth dynamics of seagrasses: A review. *J. Exp. Mar. Biol. Ecol.* 350 (1–2), 144–175. <https://doi.org/10.1016/j.jembe.2007.06.016>.
- Lovelock, C.E., Atwood, T., Baldock, J., Duarte, C.M., Hickey, S., Lavery, P.S., Masque, P., Macreadie, P.I., Ricart, A.M., Serrano, O., Steven, A., 2017. Assessing the risk of carbon dioxide emissions from large carbon ecosystems. *Front. Ecol. Environ.* 15 (5). <https://doi.org/10.1002/fee.1491>.
- Marbà, N., Duarte, C.M., 2010. Mediterranean warming triggers seagrass (*Posidonia oceanica*) shoot mortality. *Glob. Chang. Biol.* 16 (8). <https://doi.org/10.1111/j.1365-2486.2009.02130.x>.
- Marbà, N., Jordà, G., Bennett, S., Duarte, C.M., 2022. Seagrass thermal limits and vulnerability to future warming. *Front. Mar. Sci.* 9. <https://doi.org/10.3389/fmars.2022.860826>.
- Marín-Guirao, L., Ruiz, J.M., Dattolo, E., García-Munoz, R., Procaccini, G., 2016. Physiological and molecular evidence of differential short-term heat tolerance in Mediterranean seagrasses. *Sci. Rep.* 6. <https://doi.org/10.1038/srep28615>.
- Mazarrasa, I., Samper-Villarreal, J., Serrano, O., Lavery, P.S., Lovelock, C.E., Marbà, N., Duarte, C.M., Cortés, J., 2018. Habitat characteristics provide insights of carbon storage in seagrass meadows. *Mar. Pollut. Bull.* 134 (January), 106–117. <https://doi.org/10.1016/j.marpolbul.2018.01.059>.
- McHenry, J., Rassweiler, A., Hernan, G., Dubel, A.K., Curtin, C., Barzak, J., Varias, N., Lester, S.E., 2023. Geographic variation in organic carbon storage by seagrass beds. *Limnol. Oceanogr.* 68 (6). <https://doi.org/10.1002/lno.12343>.
- McMahon, K., Kilminster, K., Canto, R., Roelfsema, C., Lyons, M., Kendrick, G.A., Waycott, M., Udy, J., 2022. The risk of multiple anthropogenic and climate change threats must be considered for continental scale conservation and Management of Seagrass Habitat. *Front. Mar. Sci.* 9 (March), 1–15. <https://doi.org/10.3389/fmars.2022.837259>.
- Mtwna Nordlund, L., Koch, E.W., Barbier, E.B., Creed, J.C., 2016. Seagrass ecosystem services and their variability across genera and geographical regions. *PLoS One* 11 (10). <https://doi.org/10.1371/journal.pone.0163091>.
- Nguyen, H.M., Ralph, P.J., Marín-Guirao, L., Pernice, M., Procaccini, G., 2021. Seagrasses in an era of ocean warming: a review. *Biol. Rev.* 96 (5), 2009–2030. <https://doi.org/10.1111/bvr.12736>.
- Obura, D., Agrawal, A., DeClerck, F., Donaldson, J., Dziba, L., Emery, M.R., Friedman, K., Fromentin, J.-M., Garibaldi, L.A., Mulongoy, J., Navarrete-Frias, C., Mosig Reidl, P., Roe, D., Timoshyna, A., 2023. Prioritizing sustainable use in the Kunming-Montreal global biodiversity framework. *PLOS Sustain. Transform.* 2 (1). <https://doi.org/10.1371/journal.pstr.0000041>.
- O'Leary, J.K., Goodman, M.C., Walter, R.K., Willits, K., Pondella, D.J., Stephens, J., 2021. Effects of estuary-wide seagrass loss on fish populations. *Estuar. Coasts* 44 (8). <https://doi.org/10.1007/s12237-021-00917-2>.
- Ondiviela, B., Losada, I.J., Lara, J.L., Maza, M., Galván, C., Bouma, T.J., van Belzen, J., 2014. The role of seagrasses in coastal protection in a changing climate. *Coast. Eng.* 87. <https://doi.org/10.1016/j.coastaleng.2013.11.005>.
- Orth, R.J., Carruthers, T.J.B., Dennison, W.C., Duarte, C.M., Fourqurean, J.W., Heck, K. L., Hughes, A.R., Kendrick, G.A., Kenworthy, W.J., Olyarnik, S., Short, F.T., Waycott, M., Williams, S.L., 2006. A global crisis for seagrass ecosystems. *BioScience* 56 (12), 987–996. [https://doi.org/10.1641/0006-3568\(2006\)56\[987:AGCFSE\]2.0.CO;2](https://doi.org/10.1641/0006-3568(2006)56[987:AGCFSE]2.0.CO;2).
- Osland, M.J., Hartmann, A.M., Day, R.H., Ross, M.S., Hall, C.T., Feher, L.C., Vervaeke, W. C., 2019. Microclimate influences mangrove freeze damage: implications for range expansion in response to changing macroclimate. *Estuar. Coasts* 42 (4). <https://doi.org/10.1007/s12237-019-00533-1>.
- Pessarrodona, A., Assis, J., Filbee-Dexter, K., Burrows, M.T., Gattuso, J.P., Duarte, C.M., Krause-Jensen, D., Moore, P.J., Smale, D.A., Wernberg, T., 2022. Global seaweed productivity. *Sci. Adv.* 8 (37). <https://doi.org/10.1126/sciadv.abn2465>.
- Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Aroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., & Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. In *Ecography* (Vol. 40, issue 8). doi:<https://doi.org/10.1111/ecog.02881>.

- R Core Team, 2023. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>.
- Roca, G., Alcoverro, T., Krause-Jensen, D., Balsby, T.J.S., Van Katwijk, M.M., Marbà, N., Santos, R., Arthur, R., Mascaró, O., Fernández-Torquemada, Y., Pérez, M., Duarte, C. M., Romero, J., 2016. Response of seagrass indicators to shifts in environmental stressors: A global review and management synthesis. *Ecol. Indic.* 63. <https://doi.org/10.1016/j.ecolind.2015.12.007>.
- Schuerch, M., Spencer, T., Temmerman, S., Kirwan, M.L., Wolff, C., Lincke, D., McOwen, C.J., Pickering, M.D., Reef, R., Vafeidis, A.T., Hinkel, J., Nicholls, R.J., Brown, S., 2018. Future response of global coastal wetlands to sea-level rise. *Nature* 561 (7722), 231–234. <https://doi.org/10.1038/s41586-018-0476-5>.
- Serrano, O., Arias-Ortiz, A., Duarte, C.M., Kendrick, G.A., Lavery, P.S., 2021. Impact of marine heatwaves on seagrass ecosystems. In: *Ecosystem collapse and climate change*. Springer International Publishing, Cham, pp. 345–364.
- Short, F.T., Koch, E.W., Creed, J.C., Magalhães, K.M., Fernandez, E., Gaeckle, J.L., 2006. SeagrassNet monitoring across the Americas: case studies of seagrass decline. *Mar. Ecol.* 27 (4), 277–289. <https://doi.org/10.1111/j.1439-0485.2006.00095.x>.
- Short, F.T., Polidoro, B., Livingstone, S.R., Carpenter, K.E., Bandeira, S., Bujang, J.S., Calumpong, H.P., Carruthers, T.J.B., Coles, R.G., Dennison, W.C., Erftemeijer, P.L.A., Fortes, M.D., Freeman, A.S., Jagtap, T.G., Kamal, A.H.M., Kendrick, G.A., Judson Kenworthy, W., La Nafie, Y.A., Nasution, I.M., Zieman, J.C., 2011. Extinction risk assessment of the world's seagrass species. *Biol. Conserv.* 144 (7), 1961–1971. <https://doi.org/10.1016/j.biocon.2011.04.010>.
- Spalding, M.D., Fox, H.E., Allen, G.R., Davidson, N., Ferdaña, Z.A., Finlayson, M.A.X., Robertson, J., 2007. Marine ecoregions of the world: a bioregionalization of coastal and shelf areas. *BioScience* 57 (7), 573–583.
- Steiner, S.C.C., Willette, D.A., 2015. The expansion of *Halophila stipulacea* (Hydrocharitaceae, Angiospermae) is changing the seagrass landscape in the Commonwealth of Dominica lesser Antilles. *Caribbean Naturalist* 22 (January 2016).
- Strydom, S., Murray, K., Wilson, S., Huntley, B., Rule, M., Heithaus, M., Bessey, C., Kendrick, G.A., Burkholder, D., Fraser, M.W., Zdunick, K., 2020. Too hot to handle: unprecedented seagrass death driven by marine heatwave in a world heritage area. *Glob. Chang. Biol.* 26 (6), 3525–3538. <https://doi.org/10.1111/gcb.15065>.
- Taillardat, P., Friess, D.A., Lupascu, M., 2018. Mangrove blue carbon strategies for climate change mitigation are most effective at the national scale. *Biol. Lett.* 14 (10). <https://doi.org/10.1098/rsbl.2018.0251>.
- Tang, K.H.D., Hadibarata, T., 2022. Seagrass meadows under the changing climate: A review of the impacts of climate stressors. *Res. Ecol.* 4 (1). <https://doi.org/10.30564/re.v4i1.4363>.
- Touchette, B.W., 2007. Seagrass-salinity interactions: physiological mechanisms used by submersed marine angiosperms for a life at sea. *J. Exp. Mar. Biol. Ecol.* 350 (1–2). <https://doi.org/10.1016/j.jembe.2007.05.037>.
- Touchette, B.W., Burkholder, J.A.M., 2000. Review of nitrogen and phosphorus metabolism in seagrasses. *J. Exp. Mar. Biol. Ecol.* 250 (1–2), 133–167. [https://doi.org/10.1016/S0022-0981\(00\)00195-7](https://doi.org/10.1016/S0022-0981(00)00195-7).
- Turschwell, M.P., Connolly, R.M., Dunic, J.C., Sievers, M., Buelow, C.A., Pearson, R.M., Tulloch, V.J.D., Cote, I.M., Unsworth, R.K.F., Collier, C.J., Brown, C.J., 2021. Anthropogenic pressures and life history predict trajectories of seagrass meadow extent at a global scale. *Proc. Natl. Acad. Sci. USA* 118 (45). <https://doi.org/10.1073/pnas.2110802118>.
- Uhrin, A.V., Turner, M.G., 2018. Physical drivers of seagrass spatial configuration: the role of thresholds. *Landsc. Ecol.* 33 (12), 2253–2272. <https://doi.org/10.1007/s10980-018-0739-4>.
- Unsworth, R.K.F., Cullen-Unsworth, L.C., 2014. Biodiversity, ecosystem services, and the conservation of seagrass meadows. *Coast. Conserv.* <https://doi.org/10.1017/cbo9781139137089.005>.
- Unsworth, R.K.F., McKenzie, L.J., Collier, C.J., Cullen-Unsworth, L.C., Duarte, C.M., Eklöf, J.S., Jarvis, J.C., Jones, B.L., Nordlund, L.M., 2019. Global challenges for seagrass conservation. *Ambio* 48 (8). <https://doi.org/10.1007/s13280-018-1115-y>.
- Valavi, R., Elith, J., Lahoz-Monfort, J.J., Guillerá-Arroita, G., 2019. blockCV: an R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods Ecol. Evol.* 10 (2). <https://doi.org/10.1111/2041-210X.13107>.
- Waycott, M., Duarte, C.M., Carruthers, T.J.B., Orth, R.J., Dennison, W.C., Olyarnik, S., Calladine, A., Fourqurean, J.W., Heck, K.L., Hughes, A.R., Kendrick, G.A., Kenworthy, W.J., Short, F.T., Williams, S.L., 2009. Accelerating loss of seagrasses across the globe threatens coastal ecosystems. *Proc. Natl. Acad. Sci. USA* 106 (30), 12377–12381. <https://doi.org/10.1073/pnas.0905620106>.
- Wedding, L.M., Moritsch, M., Verutes, G., Arkema, K., Hartge, E., Reiblich, J., Douglass, J., Taylor, S., Strong, A.L., 2021. Incorporating blue carbon sequestration benefits into sub-national climate policies. *Glob. Environ. Chang.* 69. <https://doi.org/10.1016/j.gloenvcha.2020.102206>.
- Zurell, D., Elith, J., Schröder, B., 2012. Predicting to new environments: tools for visualizing model behaviour and impacts on mapped distributions. *Divers. Distrib.* 18 (6), 628–634. <https://doi.org/10.1111/j.1472-4642.2012.00887.x>.
- Strydom, S., Webster, C. L., Dea, C. M. O., Said, N. E., McCallum, R., Salinas, C., Billingham, S., Lafratta, A., Phelps, C. M., Gorham, C., Dunham, N., Bernasconi, R., Frouws, A. M., Werner, A., Vitelli, F., Puigcorbè, V., Cruz, A. D., McMahon, K. M., Robinson, J., ... Ave, D. P. (2022). Global dataset on seagrass meadow structure, biomass, production and reproduction. July, 1–17.